

FW1 ACL (Outside DMZ):

DENY: RFC1918 spoofing (10.0.x.x, 172.16.x.x, 192.168.x.x) PERMIT: TCP/80 (HTTP)
PERMIT: TCP/443 (HTTPS) PERMIT: Established TCP connections DENY: Everything else (implicit)

FW2 ACL (Inside DMZ):

PERMIT: VLAN 20 (10.0.20.x) Database (10.0.30.x) on port 3306 (MySQL) PERMIT:
VLAN 50 (10.0.50.x) Database (10.0.30.x) on port 3306 DENY: Everything else

HSRP Configuration (Example for VLAN 10):

Core-SW1: standby 1 ip 10.0.10.254, priority 110, preempt Core-SW2: standby 1 ip
10.0.10.254, priority 100, preempt

Port Assignments:

ACC-SW1: FE0/3 WEB-BLADE-1 (VLAN 20) FE0/4 WEB-BLADE-2 (VLAN 20)

ACC-SW2: FE0/3 WEB-BLADE-3 (VLAN 20) FE0/4 DB-BLADE-1 (VLAN 30)

ACC-SW3: FE0/3 DB-BLADE-2 (VLAN 30) FE0/4 VIRT-HOST-1 (VLAN 40)

ACC-SW4: FE0/3 VIRT-HOST-2 (VLAN 40) FE0/4 DNS-DHCP-NTP (VLAN 40)

ACC-SW5: FE0/3 BACKUP-1 (VLAN 70) FE0/4 BACKUP-2 (VLAN 70)

ACC-SW6: FE0/3 LOAD-BALANCER (VLAN 20) FE0/4 SOC-SERVER (VLAN 50) FE0/5
SIEM-SERVER (VLAN 60) FE0/6 IDS-SENSOR (VLAN 99) FE0/7 IPS-SENSOR (VLAN
99)

OSPF (Internal Dynamic Routing):

Process 1, Area 0 (backbone only) MD5 Authentication: BANKOSPF2026

Router IDs:

- CORE-R1: 1.1.1.3
- CORE-R2: 2.2.2.3
- CORE-SW1: 3.3.3.1
- CORE-SW2: 3.3.3.2
- AGG-SW1: 5.5.5.1

- AGG-SW2: 5.5.5.2

Network Statement:

- All routers/L3 switches: network 10.0.0.0 0.0.255.255 area 0
- This advertises all 10.0.x.x subnets

Passive Interfaces:

- VLAN SVIs on all switches (don't send hellos to end devices)

BGP (External Internet Routing):

AS 65001 (Your Bank) Neighbors:

- ISP-1 (203.0.113.1, AS 100)
- ISP-2 (203.0.113.5, AS 200)

Router IDs: Same as OSPF (1.1.1.3, 2.2.2.3)

Redistribution:

- Redistribute OSPF routes (so ISPs know how to reach 10.0.x.x)
- Redistribute connected networks (203.0.113.0/29)

Result:

- Your bank is reachable from internet via two ISP links
- Automatic failover if one ISP link fails
- Load balancing across both ISPs

SECURITY ARCHITECTURE:

Layered Defense (Defense in Depth):

Firewall Layer 1 (FW1): Denies RFC1918 spoofing Allows only HTTP/443
Stateful inspection

Firewall Layer 2 (FW2): Restricts database access Only VLAN 20/50 30:3306
Blocks lateral movement

VLAN Isolation: VLAN 20 (web) ≠ VLAN 30 (database) VLAN 40 (internal) ≠ VLAN
50 (security) Security by separation

Port Security: Unused ports shutdown (VLAN 99) IDS/IPS on separate sensors
SOC monitoring 24/7

Scenario: Customer accesses website at 203.0.113.150:

1. Internet packet arrives at FW1
 - Source: 8.8.8.8:1000, Dest: 203.0.113.150:443
2. FW1 ACL checks:
 - RFC1918? No
 - Port 443? Yes
 - PERMIT forward
3. NAT Translation (CORE-R1 or R2):
 - 203.0.113.150 10.0.20.100 (Load-Balancer)
 - Forward to load-balancer
4. VLAN 20 routing:
 - Packet: Dest 10.0.20.100
 - Router: Check routing table
 - Next hop: 10.0.20.1 (CORE-SW1 or SW2 via HSRP)
5. CORE-SW1/2 routing:
 - Destination on same VLAN
 - Forward to access switch (ACC-SW6)
6. Access switch forwarding:
 - Port assignment: Load-Balancer on FE0/3
 - Frame arrives at Load-Balancer
7. Load-Balancer processes:
 - Distributes to WEB-BLADE-1, 2, or 3
 - Web server responds
8. Return path (automatic):

- Reverse routing through same path
- NAT translates back: 10.0.20.100 → 203.0.113.150
- FW1 allows established connection
- Response reaches internet customer

Total latency: 5-20ms depending on device load